

SIMATS ENGINEERING

P.L.OAVIKSHA
(192572001)

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA1715

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

Duration :1 Hour

QUESTIONS: 4

Marks:40

Total

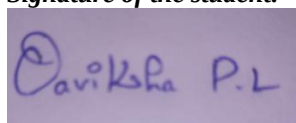
Annexure A

CASE STUDY

Code of Conduct:

I _P.L.OAVIKSHA(192572001) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- ii. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- iii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ASSESSMENT TOOL 1 – ARTIFICIAL INTELLIGENCE

CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

(a) Representation using Propositional Logic

Let:

- **F** = Patient has Fever
- **C** = Patient has Cough
- **R** = Patient has Rash
- **B** = Patient has Breathlessness
- **FL** = Possible Flu
- **M** = Possible Measles
- **P** = Possible Pneumonia

Rules

Rules	
R1:	$F \wedge C \rightarrow FL$
R2:	$F \wedge R \rightarrow M$
R3:	$C \wedge B \rightarrow P$

Facts of Patient A

[F = True]

[C = True]

[B = True]

Therefore, the knowledge base is:

[KB={F,C,B,(F $\wedge$ C $\rightarrow$ FL),(F $\wedge$ R $\rightarrow$ M),(C $\wedge$ B $\rightarrow$ P)}]

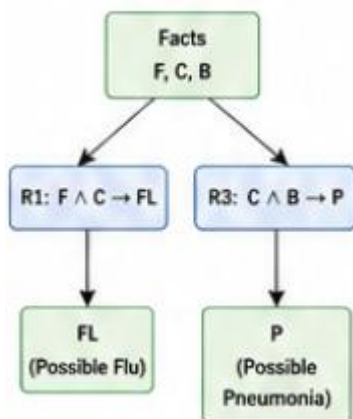
(b) Forward Chaining

Step	Known Facts	Rule Fired	New Fact Derived	
0	F, C, B	–	–	Initial facts
1	F, C, B	R1: $F \wedge C \rightarrow FL$	FL	F and C are true
2	F, C, B, FL	R2: $F \wedge R \rightarrow M$	–	R is not known
3	F, C, B, FL	R3: $C \wedge B \rightarrow P$	P	C and B are true

Final Derived Facts: Possible Flu (FL), Possible Pneumonia (P)

Forward-Chaining Result

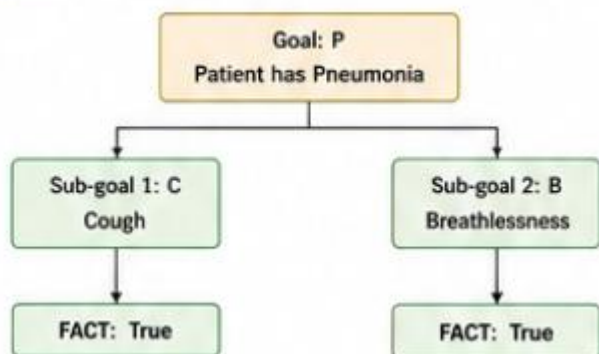
Forward Chaining Diagram



(c) Backward Chaining for “Patient A has Pneumonia”

Goal:

Rule used: R3: $C \wedge B \rightarrow P$



Conclusion: The goal is successfully proved by backward chaining.

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

The case study provides three FOL rules and facts concerning an ambulance and CarA.

(a) Unification

Given Rules (FOL)

R1: $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

R2: $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

R3: $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

Facts

- $\text{Vehicle}(\text{Ambulance})$
- $\text{EmergencyType}(\text{Ambulance})$
- $\text{Vehicle}(\text{CarA})$
- $\text{Behind}(\text{CarA}, \text{Ambulance})$

Unification Steps

Step	Rule Expression	Fact	Substitution
1	$\text{Vehicle}(x)$	$\text{Vehicle}(\text{Ambulance})$	$x/\text{Ambulance}$
2	$\text{EmergencyType}(x)$	$\text{EmergencyType}(\text{Ambulance})$	$x/\text{Ambulance}$
3	$\text{ClearPath}(x)$	$\text{ClearPath}(\text{Ambulance})$	$x/\text{Ambulance}$

(b) Resolution Proof that $\text{Proceed}(\text{CarA})$ is True

Step 1 – Convert Rule 1 to CNF

Convert Rules to CNF

C1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

C2: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

C3: $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

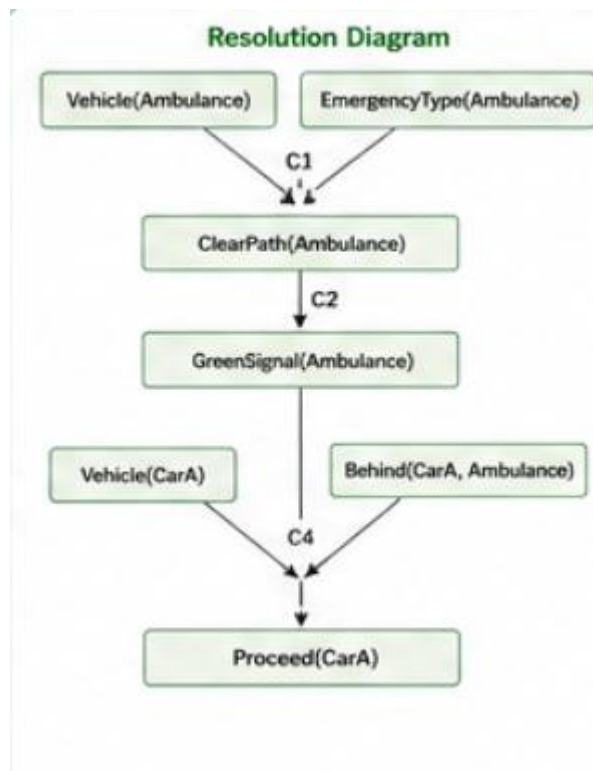
C4: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Resolution Steps

1. From C1 with $\text{Vehicle}(\text{Ambulance})$ and $\text{EmergencyType}(\text{Ambulance})$
 $\Rightarrow \text{ClearPath}(\text{Ambulance})$
2. From C2 with $\text{ClearPath}(\text{Ambulance})$
 $\Rightarrow \text{GreenSignal}(\text{Ambulance})$
3. From C4 with $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$, $\text{GreenSignal}(\text{Ambulance}) \Rightarrow \text{Proceed}(\text{CarA})$

Hence, $\text{Proceed}(\text{CarA})$ is **TRUE**.

Resolution Proof Diagram



(c) Two Simultaneous Emergency Vehicles

The current knowledge base can process multiple vehicles because the rules use the variable **x**.

For example, if a second ambulance exists:

[Vehicle(Ambulance2)]

[EmergencyType(Ambulance2)]

Rule 1 derives:

[ClearPath(Ambulance2)]

Rule 2 derives:

[GreenSignal(Ambulance2)]

However, the current KB does not contain information about:

- Priority between two emergency vehicles
- Which vehicle should receive the green signal first
- Whether two emergency vehicles can simultaneously proceed

- Conflict resolution between their routes
- Relative positions of two emergency vehicles

Additional Rules Needed

- Priority Rule: $\text{EmergencyType}(x) \wedge \text{Priority}(x, y) \rightarrow \text{GreenSignal}(x)$
- Conflict Rule: $\text{Conflict}(x, y) \wedge \text{Priority}(x, y) \rightarrow \text{Hold}(y)$

Conclusion

The existing KB supports multiple vehicles at a basic level, but it is **not sufficient for complete simultaneous emergency management** because it lacks priority and conflict-resolution knowledge.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

The knowledge base contains three implication rules and two facts.

(a) CNF Conversion

Complete CNF

Sentence	CNF
P1	$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
P2	$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
P3	$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$
Fact 1	SoilDry
Fact 2	CropWheat

Sentence	Conversion Steps	CNF
P1	$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$ $\equiv \neg \text{SoilDry} \vee \text{IrrigationNeeded}$	$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
P2	$\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$ $\equiv \neg (\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$ $\equiv \neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$	$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
P3	$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$ $\equiv \neg (\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$ $\equiv \text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$	$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$
Fact 1	SoilDry	SoilDry
Fact 2	CropWheat	CropWheat

(b) Resolution Refutation for ApplyDripMethod

Goal

Negate goal: $\neg \text{ApplyDripMethod}$

Clauses (CNF):

C1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

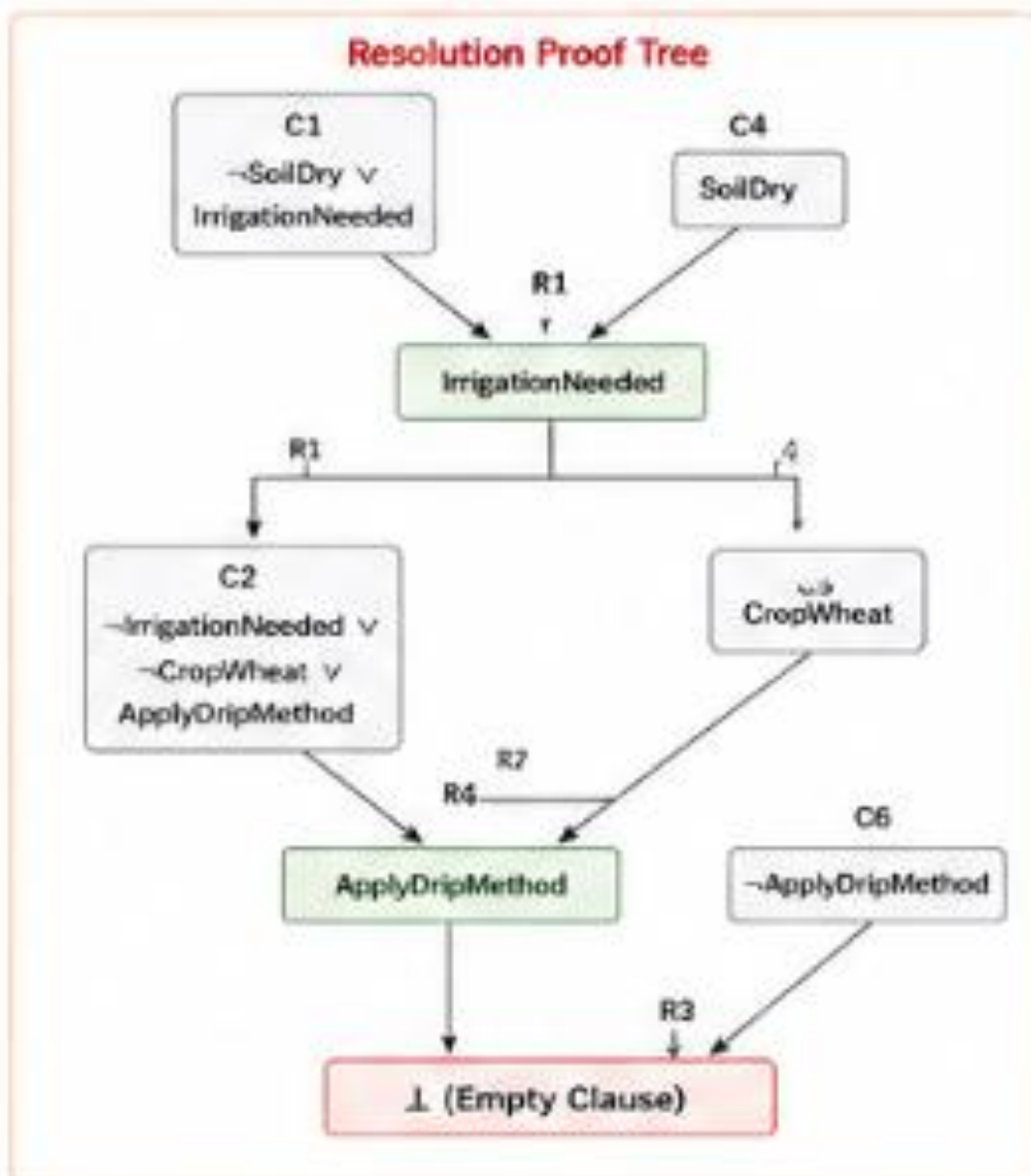
C2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

C3: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

C4: SoilDry

C5: CropWheat

C6: $\neg \text{ApplyDripMethod}$ (negated goal)



(c) Removing SoilDry

Final Result

Knowledge	Before removing SoilDry	After removing SoilDry
SoilDry	True	Unknown
IrrigationNeeded	Derived	Not derived
CropWheat	True	True
ApplyDripMethod	Derived	Not derived
CropAtRisk	Not derived	Not derived

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

The agent perceives **Stench at [1,2]**, **Breeze at [1,1]**, and **Glitter at [2,2]**. The rules in the assessment define how these percepts produce knowledge.

(a) Belief State and Modus Ponens

Given Percepts

[Stench[1,2]]

[Breeze[1,1]]

[Glitter[2,2]]

From Stench[1,2] and R 1
⇒ WumpusAdjacent[1,2]

From Breeze[1,1] and R 2
⇒ PitAdjacent[1,1]

From Glitter[2,2] and R 3
⇒ Gold[2,2]

R4 cannot be fired (no negative facts available).

Derived Beliefs:

- ✓ Wumpus is adjacent to [1,2]
- ✓ Pit is adjacent to [1,1]
- ✓ Gold is at [2,2]

(b) Forward Chaining – Possible Wumpus and Pit Locations

Assuming the usual Wumpus World adjacency relation:

From Stench[1,2]: Wumpus is adjacent to [1,2]

Adjacent cells:

[1,1]

[1,3]

[2,2]

Possible Wumpus locations

From Breeze[1,1]: Pit is adjacent to [1,1]

Adjacent cells:

[1,2]

[2,1]

Possible Pit locations

(c) Safety of Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

The proposed path is:

(c) Is path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ safe?

Cell	Knowledge about the cell	Safe?
[1,1]	Adjacent to [1,2] where stench is perceived ⇒ Possible Wumpus location	Not guaranteed
[1,2]	Adjacent to [1,1] where breeze is perceived ⇒ Possible Pit location	Not guaranteed
[2,2]	Glitter perceived ⇒ Gold is here But also adjacent to [1,2] which has stench ⇒ Possible Wumpus location	Not guaranteed

Conclusion: The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ cannot be confirmed as safe. The agent should not assume the path is safe.

Cell [1,1]

There is no direct fact saying that [1,1] contains a Wumpus or Pit.

However, from Stench[1,2], [1,1] is a possible Wumpus location.

Therefore its safety is **not guaranteed**.

Cell [1,2]

From Breeze[1,1]:

[
PitAdjacent[1,1]
]

Therefore [1,2] is a possible Pit location.

Thus [1,2] cannot be proven safe.

Cell [2,2]

The Glitter percept tells us:

[Gold[2,2]]

However, from Stench[1,2], [2,2] is also a possible Wumpus location.

Therefore [2,2] is also not proven safe.

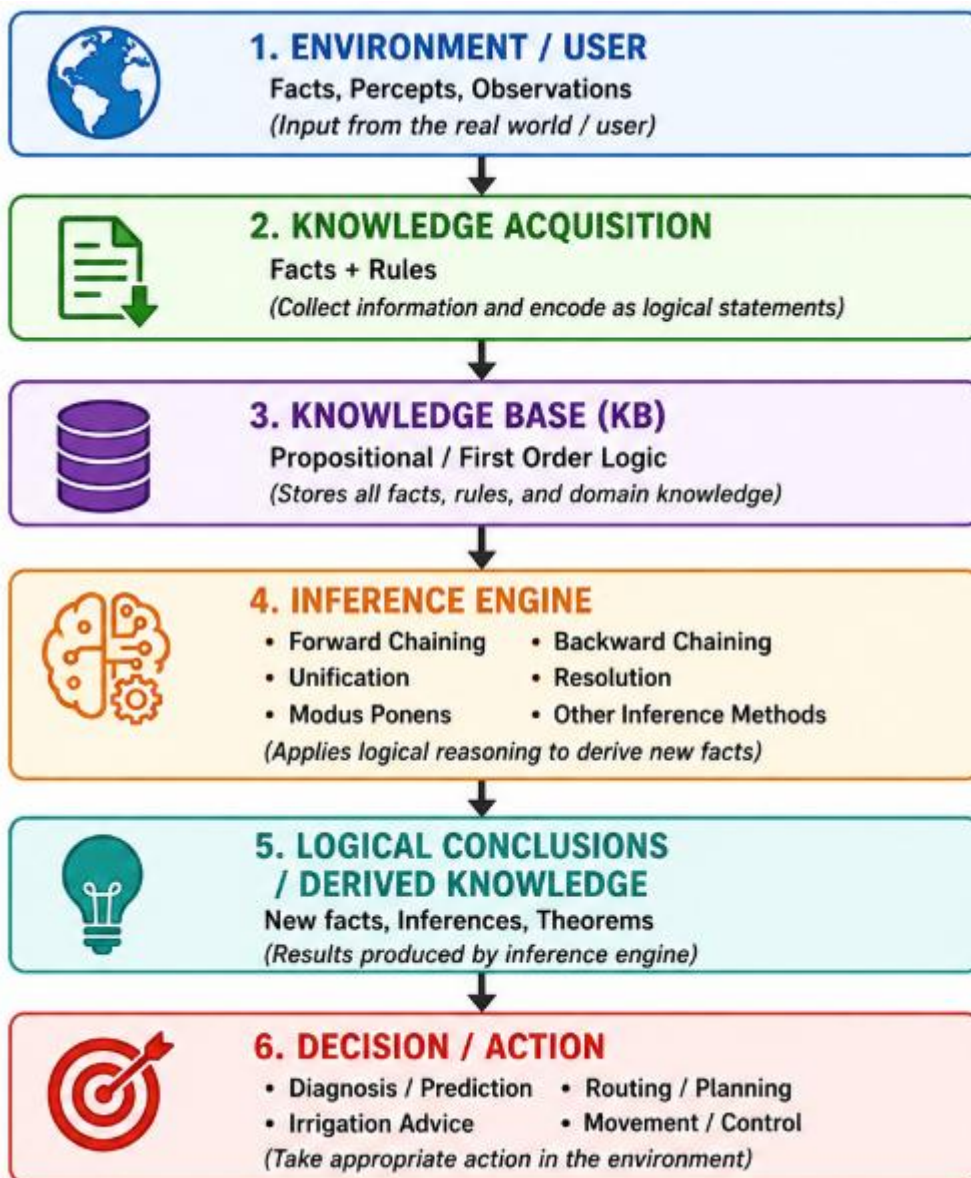
Safety Evaluation

Cell	Knowledge	Safe?
[1,1]	Possible Wumpus location	Not guaranteed
[1,2]	Possible Pit location	Not guaranteed
[2,2]	Gold present; possible Wumpus location	Not guaranteed

The agent should **not assume the path is safe** because the KB does not contain sufficient negative information to prove that these cells are free from hazards.

OVERALL ARCHITECTURE OF THE LOGICAL AI AGENT

A common architecture for the four case studies can be represented as:



Final Conclusions

Case Study	Main AI Technique	Final Result
Smart Medical Diagnosis	Forward & Backward Chaining	Possible Flu and Pneumonia; Pneumonia goal proved
Autonomous Traffic Management	Unification & Resolution	Proceed(CarA) proved true
Agricultural Expert System	CNF & Resolution Refutation	ApplyDripMethod proved
Wumpus World	Modus Ponens & Forward Chaining	Gold located at [2,2], but proposed path is not proven safe